Masterclass in Social Research

Welcome and Introduction

Plan for today

- **About Masterclass**
  - History and module objectives
  - Weekly syllabus
  - Assessment and requirements
  - Module contacts
- **Introduction**: Tell us about yourself
- **Exercise**: The mysterious case of "AI for Social Research"

History and objectives

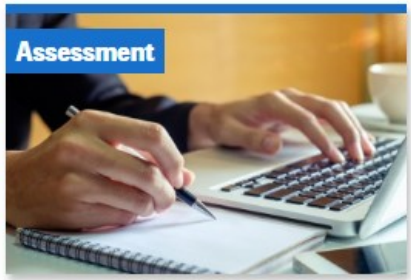
- **Masterclass in Social Research** is the final compulsory module for **MSc/PgDip Social Research** students
  - Masterclass was created at first to provide SR students with a variety of methods training
  - While serving as a compulsory module for SR students, Masterclass is also open to PGR students
  - The topics covered may change from year to year based on staff availability and new developments in social research
- Following recent program restructuring, Masterclass now serves two main objectives
  - Produce a new group of competent social researchers with different career objectives and academic ambitions
  - Showcase cutting-edge social research methods while helping students reflect on some of the most common ones

Module organization

- "Master" lecture series (6-7:30pm)
  - Each week will feature a speaker who is an expert in the selected issue and method
  - The speakers will decide how they will deliver the lectures, with the use of different pre-sessional and/or class activities
  - **The lecture series is open to postgraduate research students and their participation will not be credited**
- "Pieces-of-the-craft" workshops (7:40-9pm)
  - Each week will focus on a social research "skill" and the workshop activities will help students complete the required assessment
  - Between Weeks 6-9, students will participate in the "Social Research Dissertation Fairs," where they share their research ideas and receive feedback

Welcome!

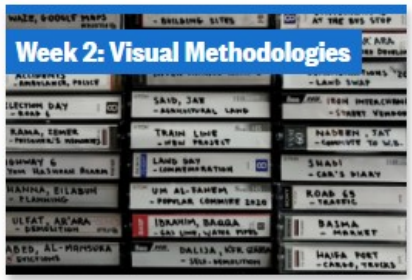
Your Studies


Assessment

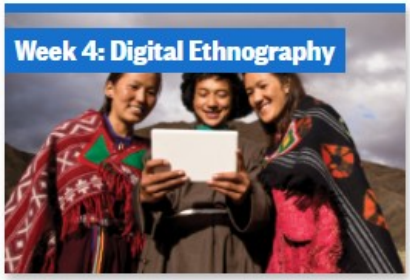
POC Workshops: AI for Social Research
Hidden from students

Dissertation Fairs

Week 1: Introduction and Welcome


Week 2: Visual Methodologies

Week 3: Surveys


Week 4: Digital Ethnography

Week 5: Oral Histories

Reading Week

Week 6: Discourse Analysis

Week 7: Computational Social Science

Week 8: History and Archives

Week 9: Mixed Methods

Week 10: Quali-Quant Methods for Social Media Research

Plan for Pieces-of-the-craft (PoC) Workshops

- **Weeks 2-4**: "AI for Social Research"
  - Week 2: Introduction
  - Week 3: AI and qualitative social research (interviews and discourse analysis)
  - Week 4: AI and Quantitative Social Research (synthetic data and agentic data analytical pipelines)
- **Week 5**: Communicating Your Research
- **Weeks 6-9**: Social Research Dissertation Fairs

Assessment (if you are taking Masterclass for credits)

- Two **blog posts** (40%): Write a blog post to summarize and reflect on **two of the lectures**; one lecture from Weeks 2-5 with the other one from Weeks 6-10
- A **critical reflection paper** (60%): (4,000 words max) to discuss how Masterclass helps you advance/rethink social research by providing
  - a summary of your research project,
  - a discussion on how AI can change/help with your project and
  - point-to-point responses to the questions and/or comments you receive during the presentation

Role of the moderator (i.e., Chao-yo)

- The moderator will join the lecture series to
  - Introduce the speaker
  - Moderate the Q&A or discussion, if necessary
- The moderator will lead the workshops and provide feedback on your assignments
- The moderator is available to answer your questions about the module
  - Students are encouraged to schedule 1:1 meetings during my office hours (Friday 2-4pm) or by appointment
  - All meetings take place via Teams by default – email if you'd like to meet in person

Notes for students who are auditing Masterclass

- Attendance at the speaker series and the "AI for Social Research" workshops is selective and entirely at your discretion
- Attendance is voluntary and will not be recorded
- Please consult your supervisor beforehand to ensure that your participation contributes meaningfully to your doctoral research
- You will be invited to contribute to a blog post at the end of the term

Knowing each other

- Please share your name, course, and subject area (e.g., political science) with us
- What are the methods you have learned or used previously?
- What project are you working on at this moment?

Recap: Investigating the social world as a vocation

- As a social researcher, we aim to produce "valid" knowledge about the social world
- The production of valid knowledge involves a series of careful thinking about ontology/epistemology, theory-building, and methodology (or the theory-empirics alignment)
  - **Reveal nuances of the real world**: constructivism and interpretivism ; theory as approach; interpretation; qualitative; inductive/abductive
  - **Search for the "ultimate" truth**: positivism; theory as paradigm; explanation; quantitative; deductive
- A new definition of "better" social research is emerging as boundaries across subject areas and research methods are constantly being contested and redefined

Table 1.1 **A Catalog of Social Science Biases**

		Threat	*Strategy to Address*	*Capacity to Address*
Individual Diversity	Demographic Biases	Limited topics and interpretations	Diversification and listening	Strong and increasing
	Geographic Biases	Limited generalization	Globalization, cross-cultural comparison	Increasing; differs by discipline
	Political Biases	Limited goals and interpretations	Empathetic recognition and sensitivity	Decreasing; recognition increasing
Scientific Practice	Methodological Biases	Incorrect findings and interpretations	Reproducibility reforms; empiricism	Strong and increasing
	Disciplinary Biases	Incomplete coverage; non-integration	Interdisciplinary research; team science	Strong and increasing
	Institutional Biases	Limited goals and topics; translation	Reform of evaluation and rewards	Strong
Innate Difficulties	Temporal Biases	Limited generalization	Long time horizons; historical data	Increasing, inherently limited
	Cognitive Biases	Confirmation bias; non-cumulation	Reproducibility reforms; meta-science studies	Increasing, inherently limited
	Application Biases	Limited goals and topics; translation	Learning from history; tracking basic science	Inherently limited

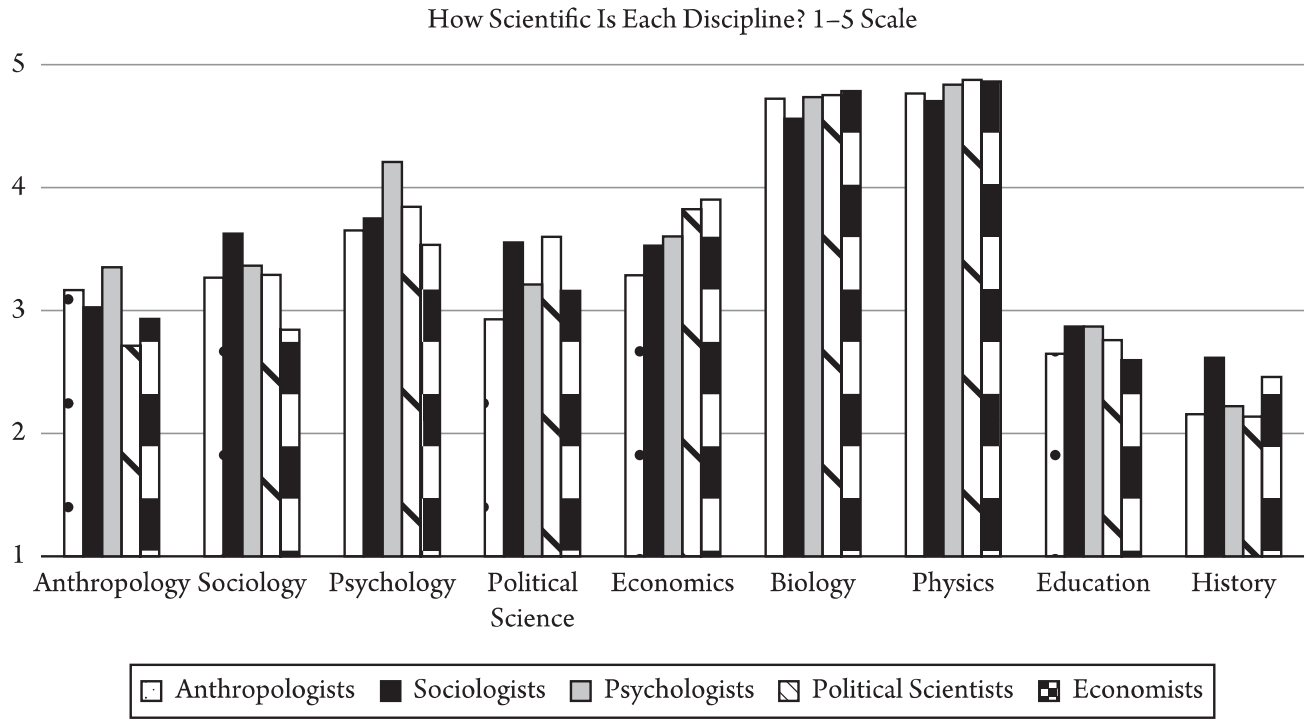
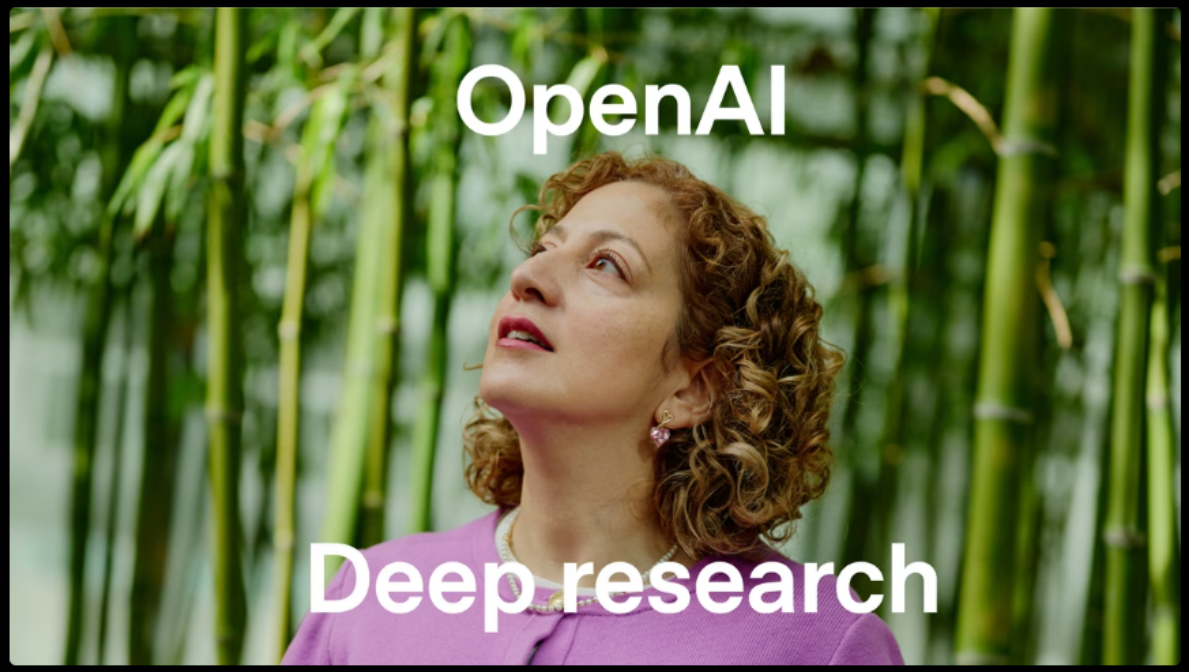


Figure 1.1 Social Scientists' Ratings of Whether Disciplines Are Scientific. Data from the author's 2020 survey of social scientists at major US research universities. n=1,141.

February 2, 2025 ChatGPT

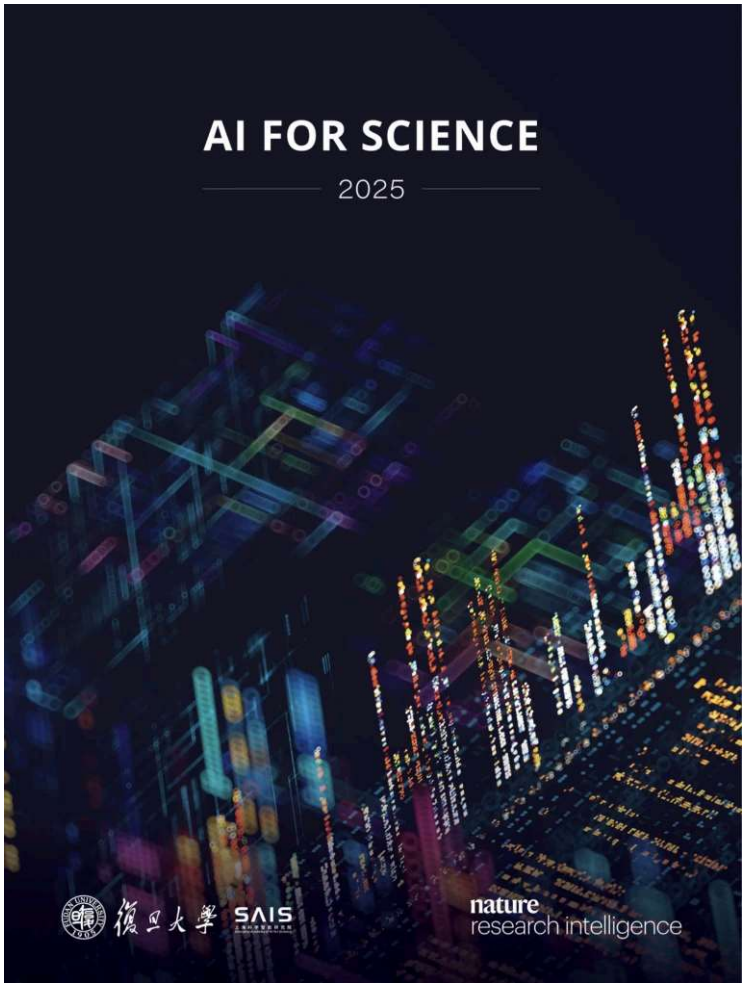
OpenAI deep research

Bain & Company researcher Reem Anchassi uses OpenAI deep research to understand complex industry trends.

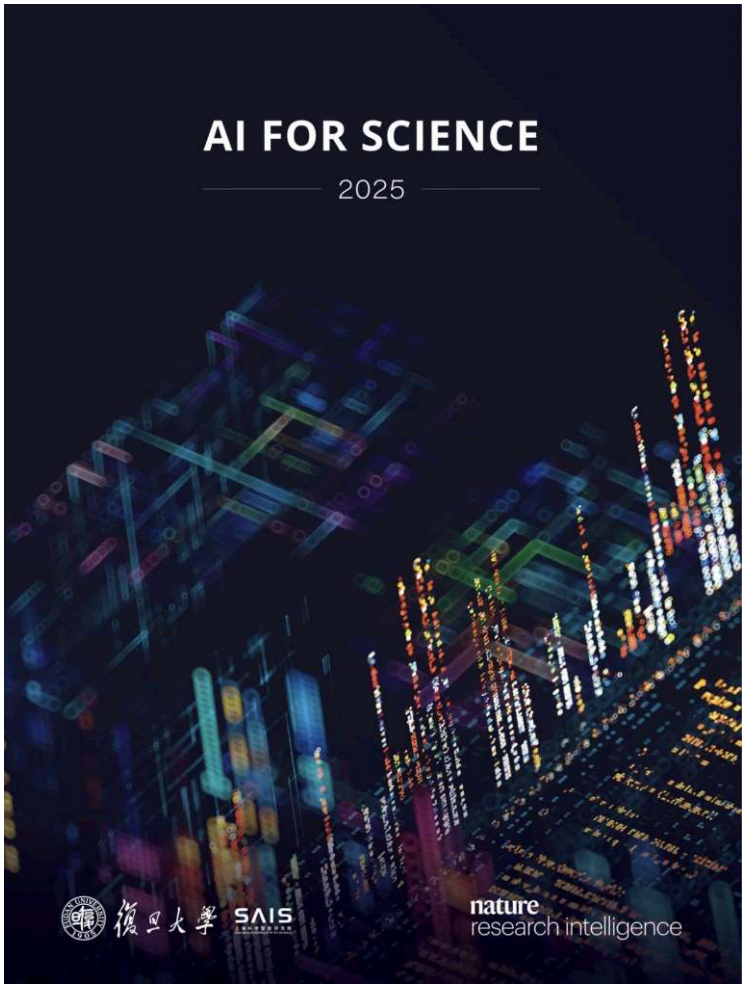


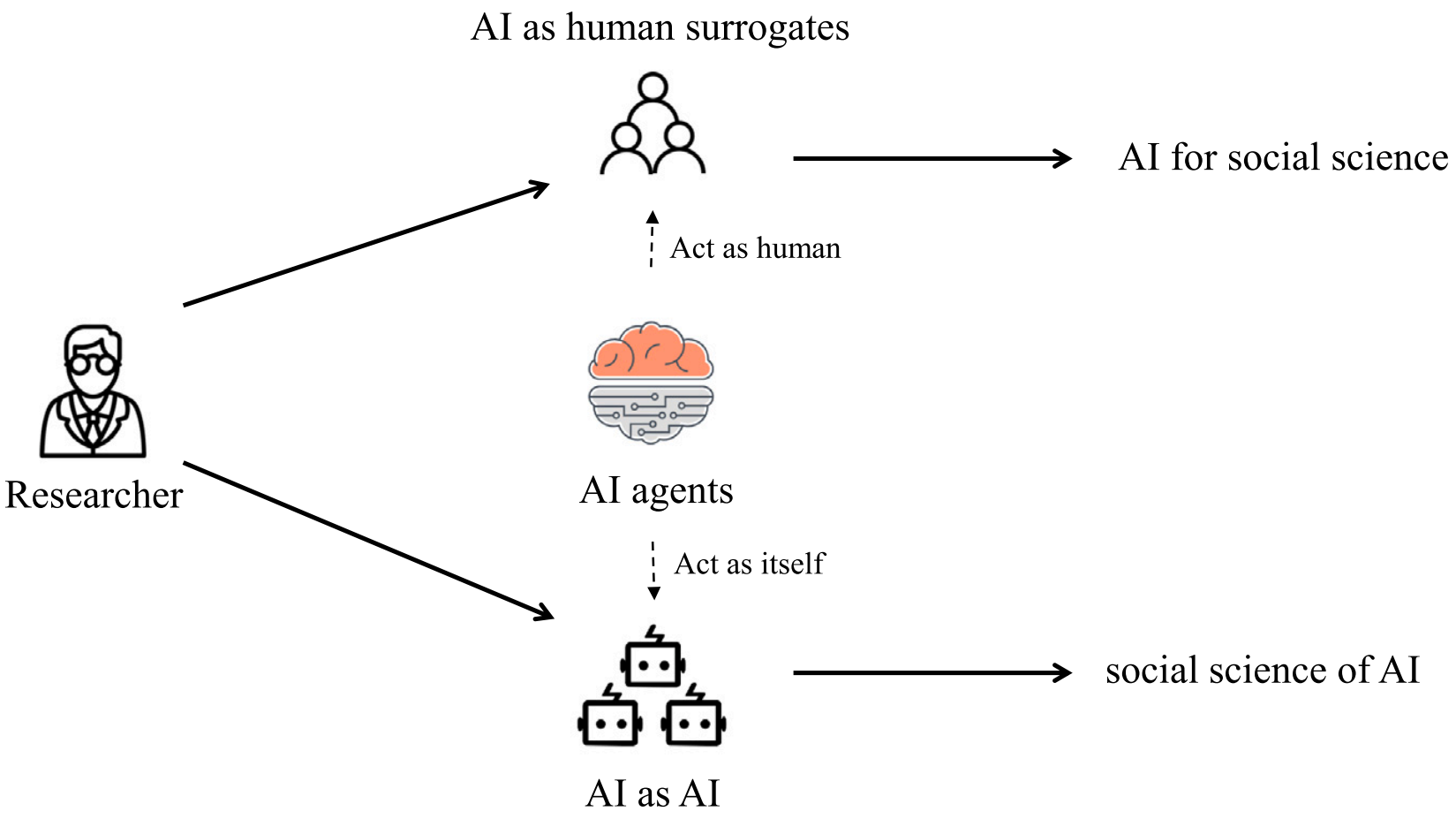
https://openai.com/research

AI-4-Science: A paradigm shift?



"AI for Science (AI4S) represents the **convergence** of artificial intelligence (AI) innovation in scientific research and AI-driven scientific discovery, demonstrating their deep integration, and the establishment of a transformative research paradigm."



"Traditional scientific discovery involves **generating and validating candidate hypotheses** from a large solution space, often characterized by low efficiency and challenges in identifying high-quality solutions. AI harnesses its **powerful data processing and analytical capabilities** to navigate solution spaces more efficiently, enabling the generation of high-quality candidate hypotheses."

AI and social sciences